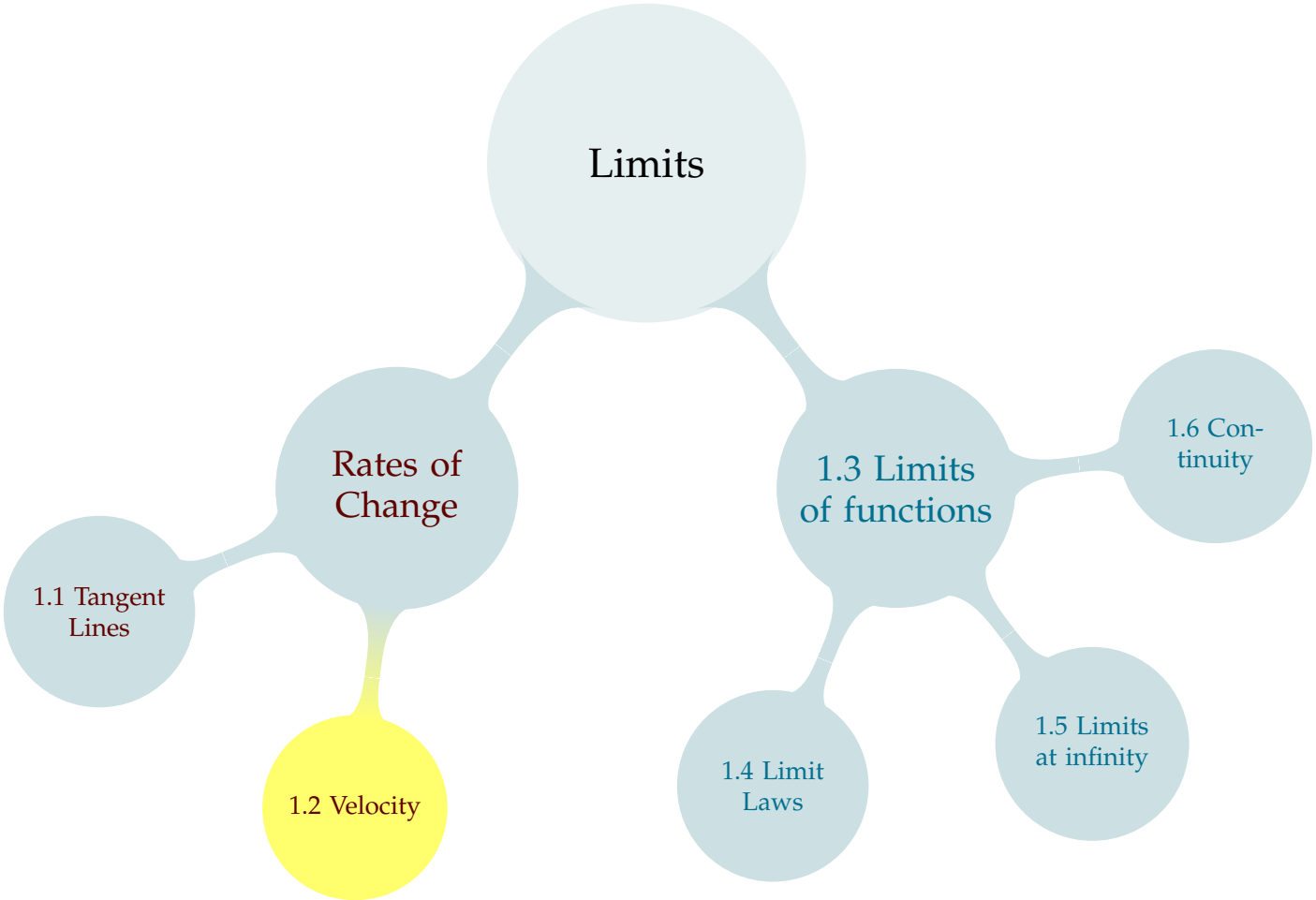
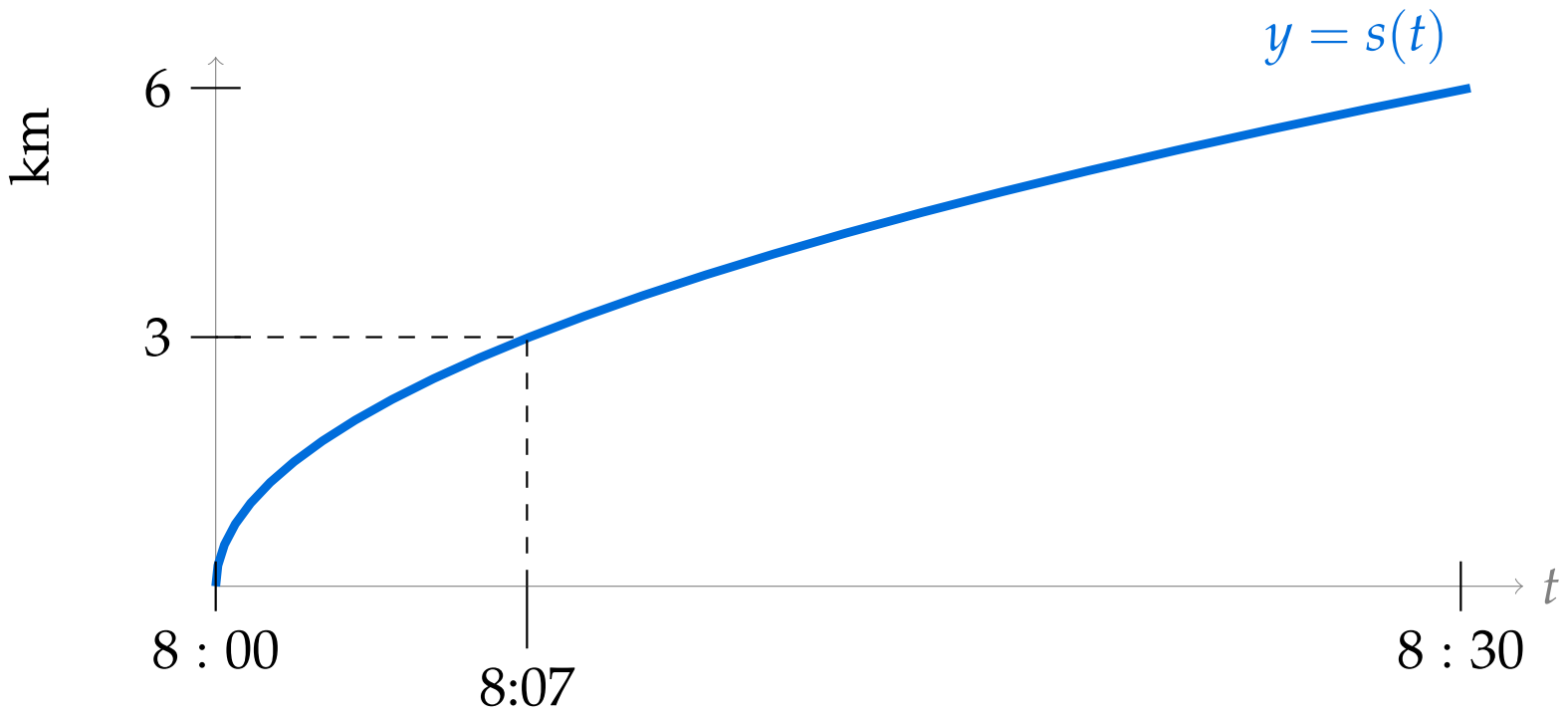
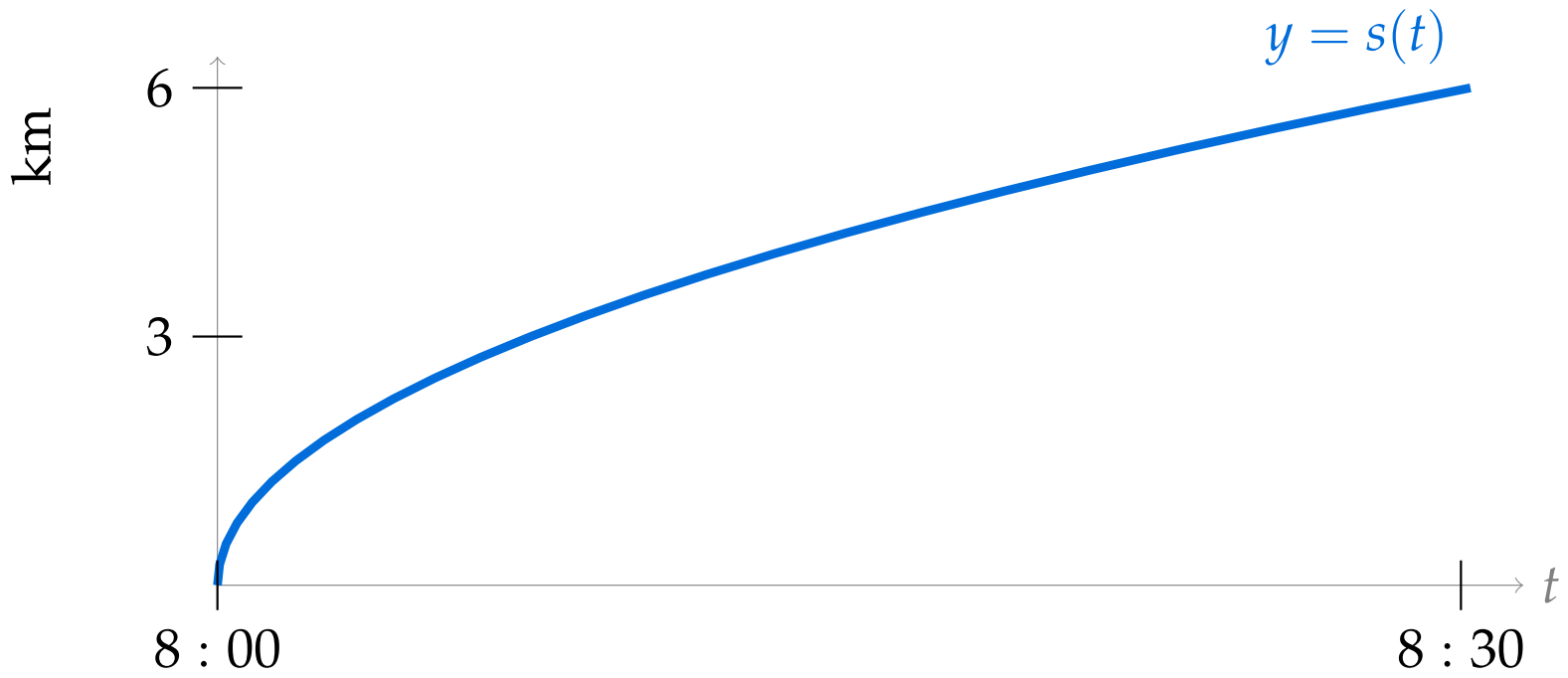
Table of Contents




$y = s(t)$
km
6
3
t
8 : 00
8 : 30


$y = s(t)$
km
6
3
t
8 : 00
8:07
8 : 30



It took $\frac{1}{2}$ hour to bike 6 km. 12 kph represents the:

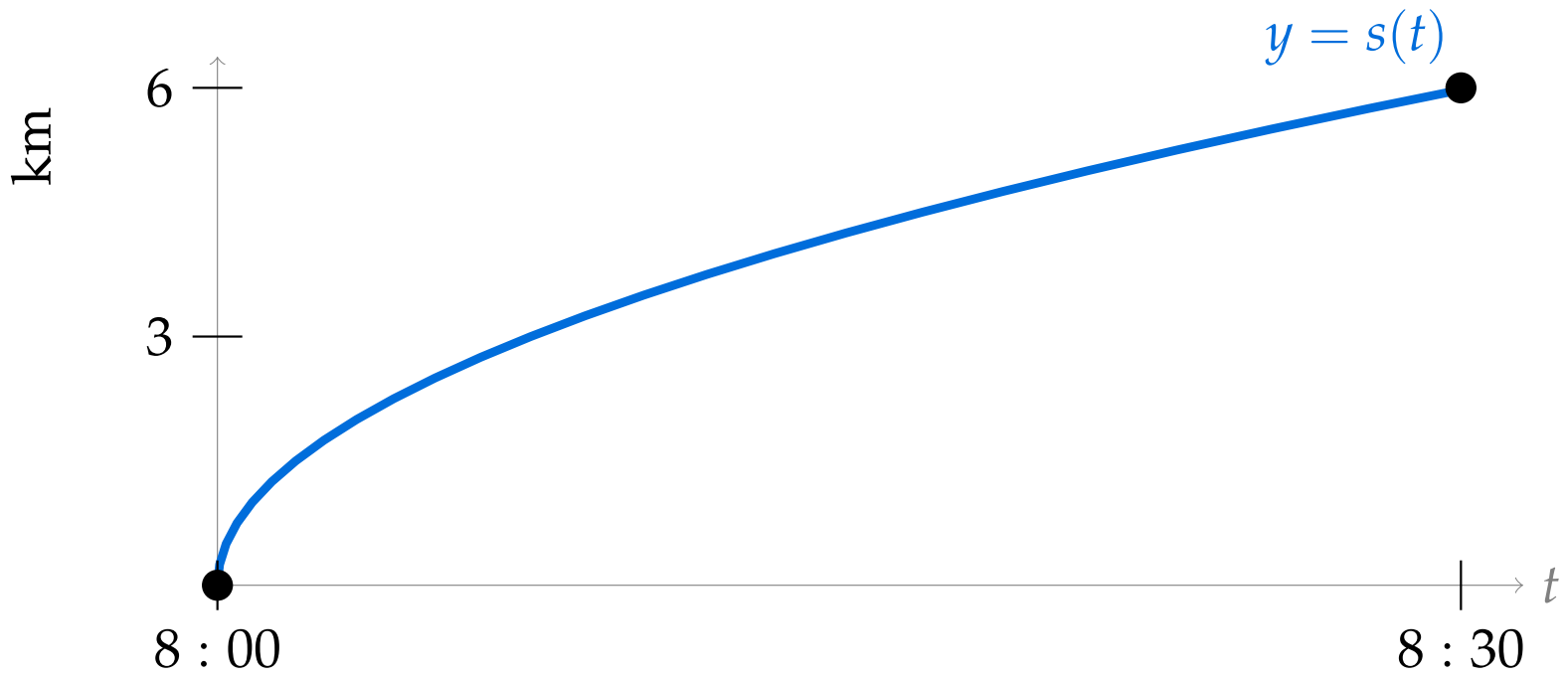
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ans



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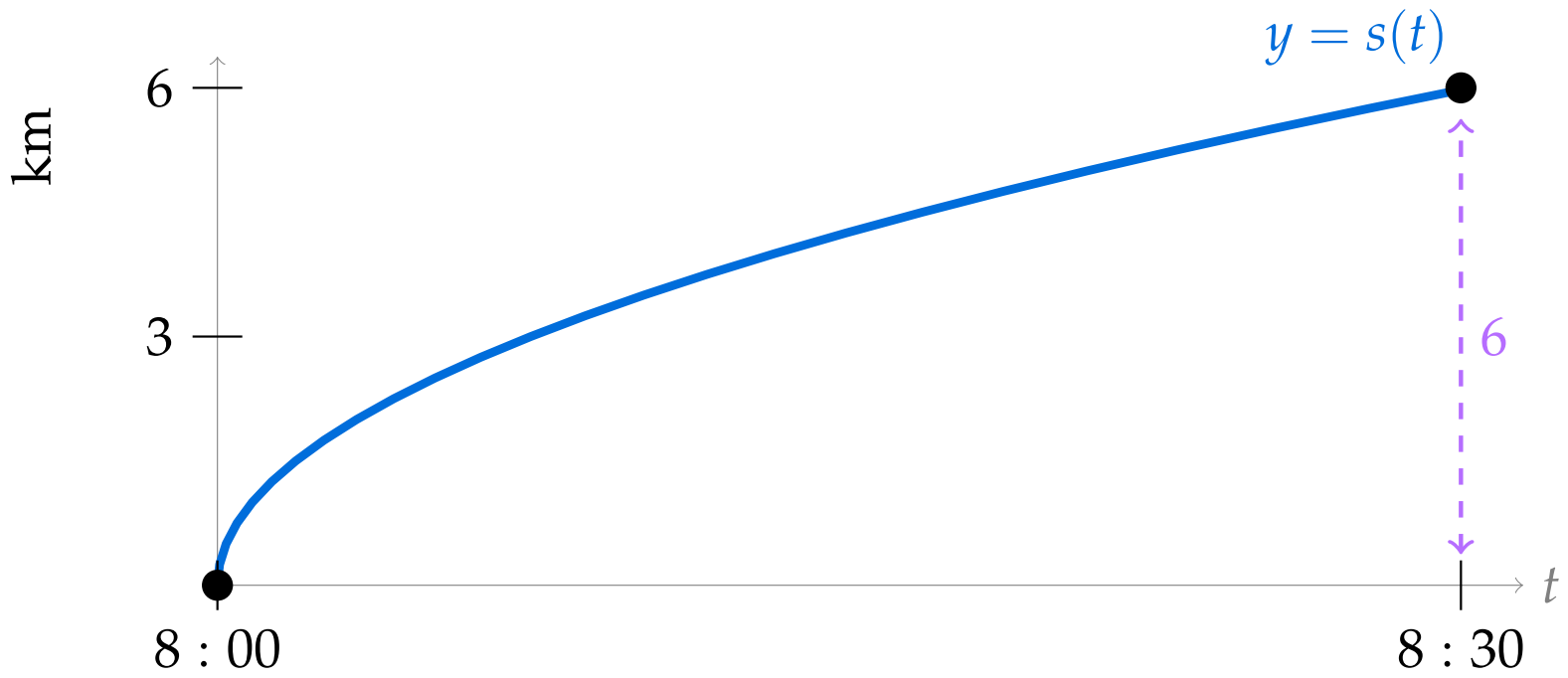
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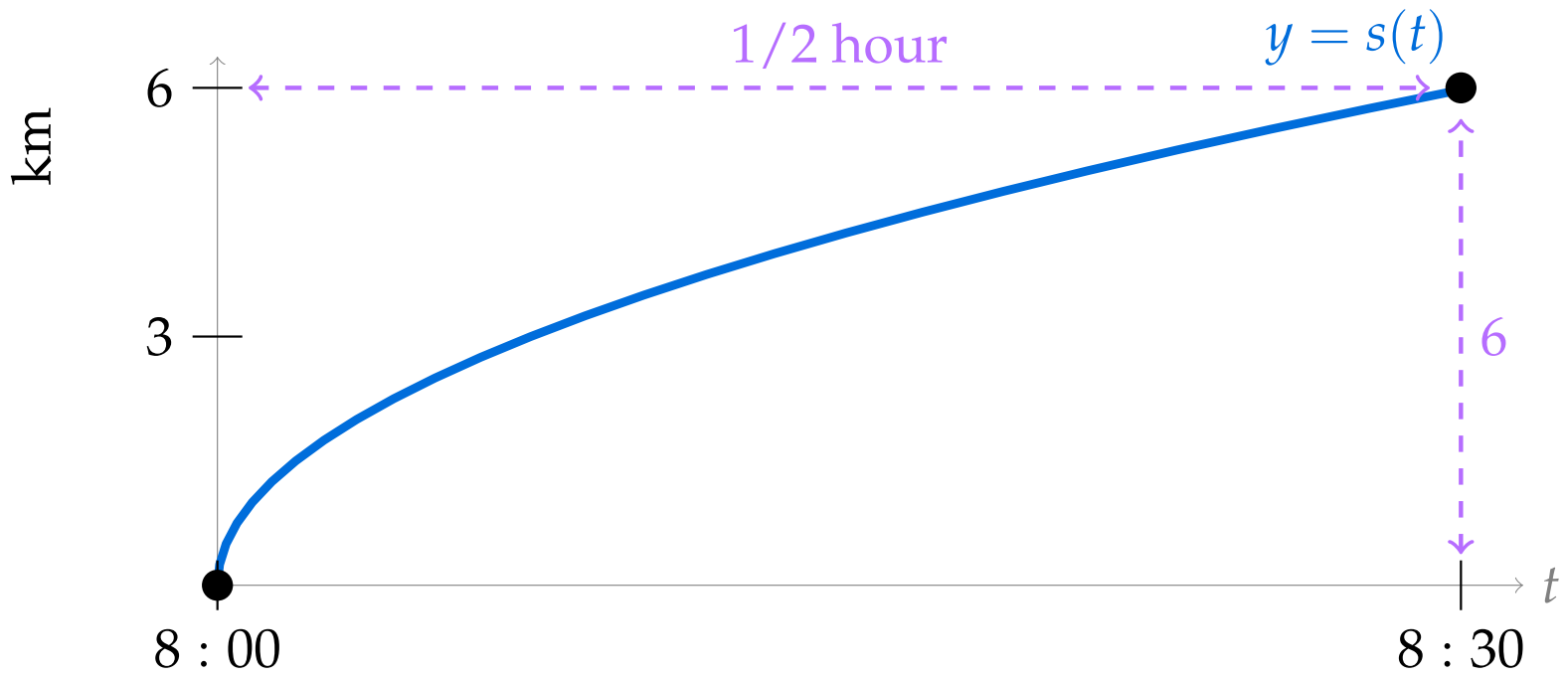
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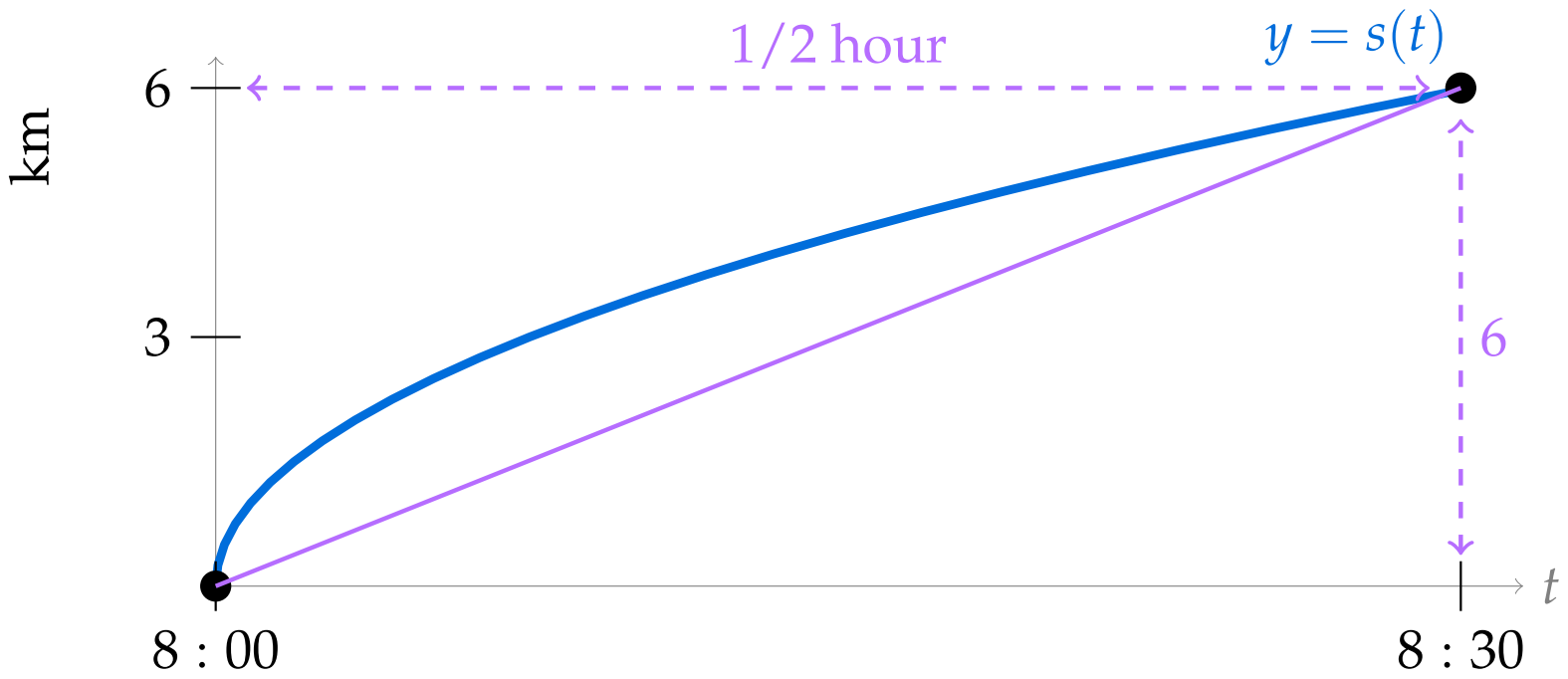
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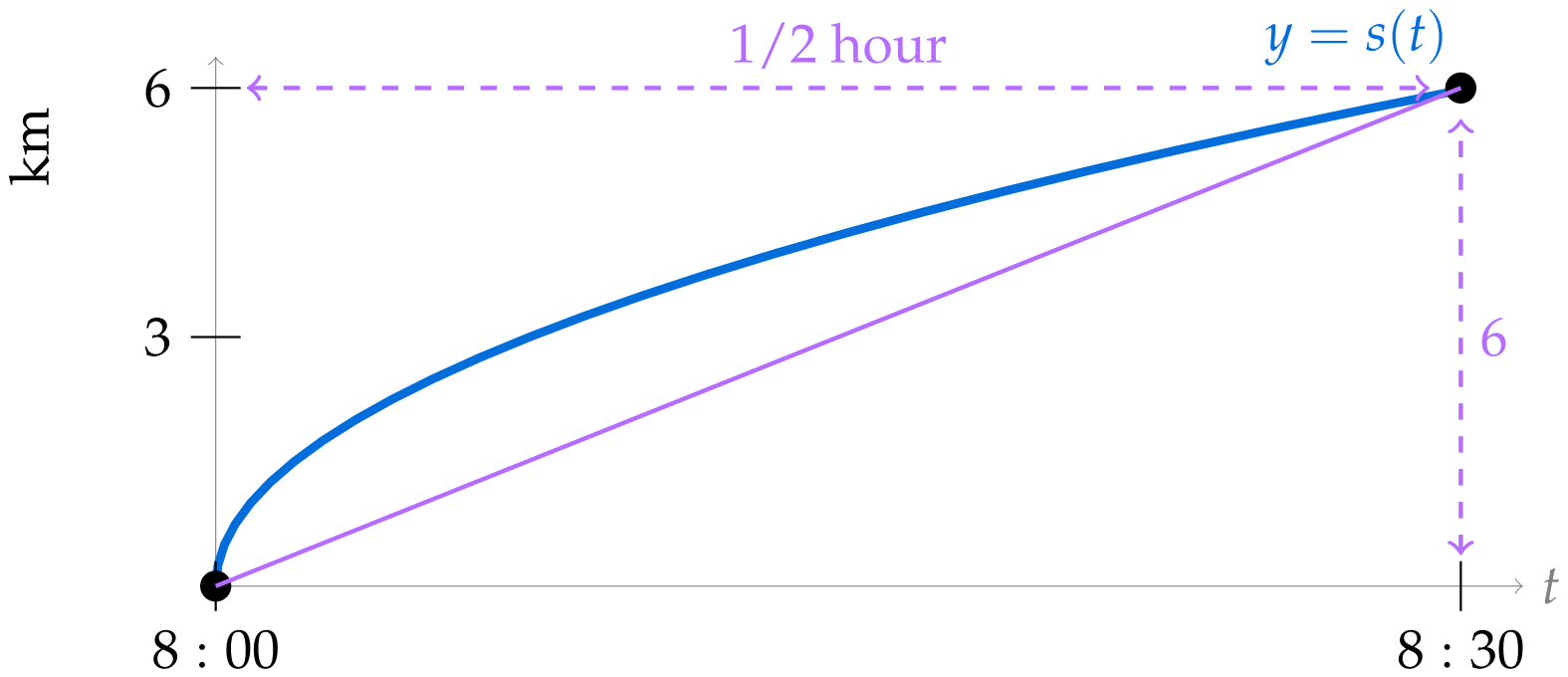
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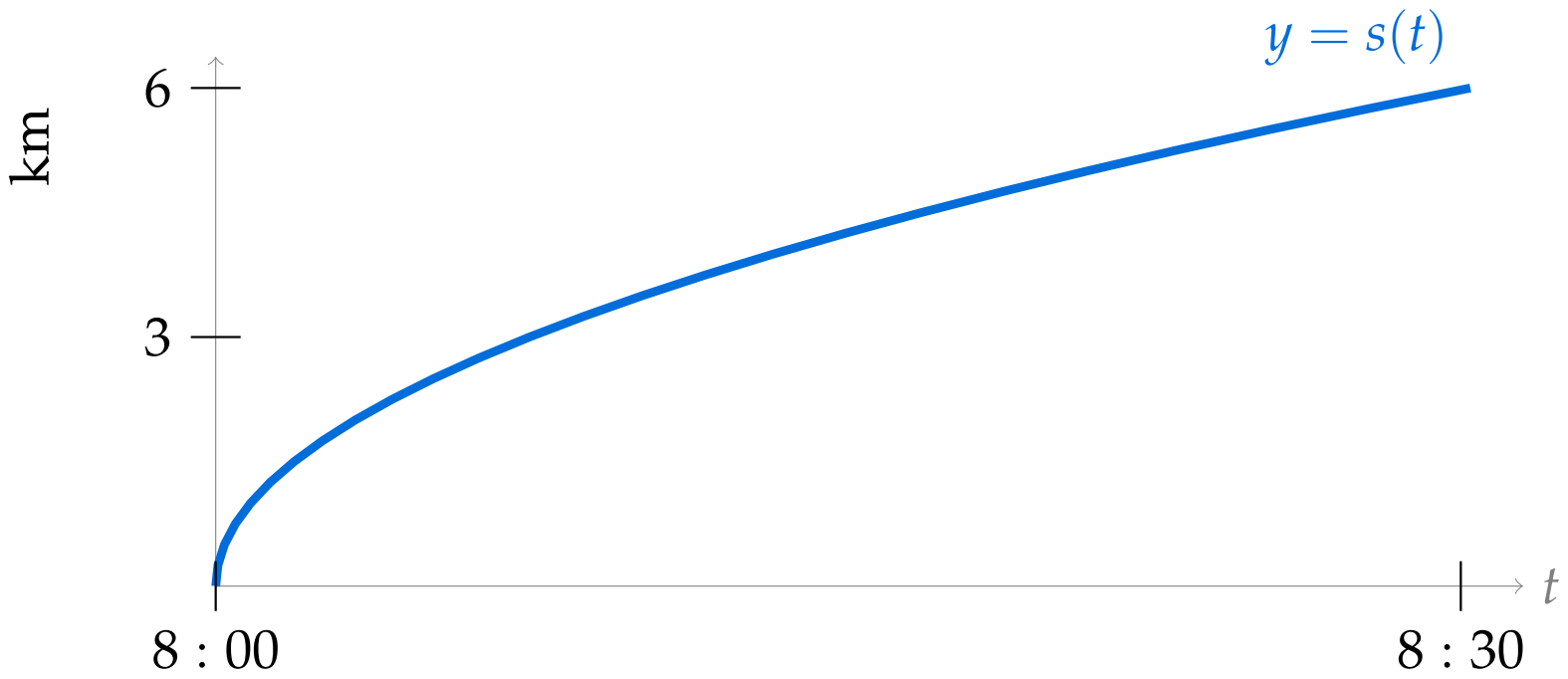
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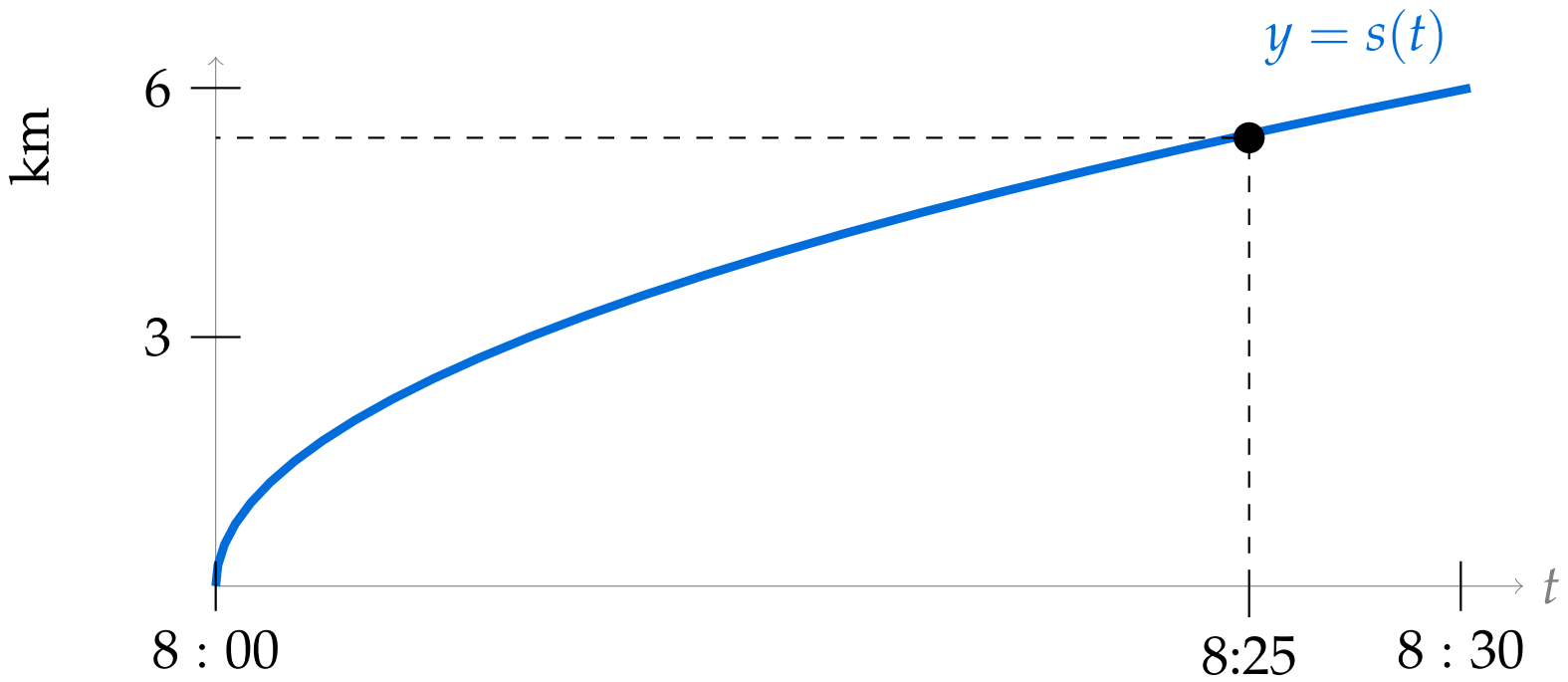
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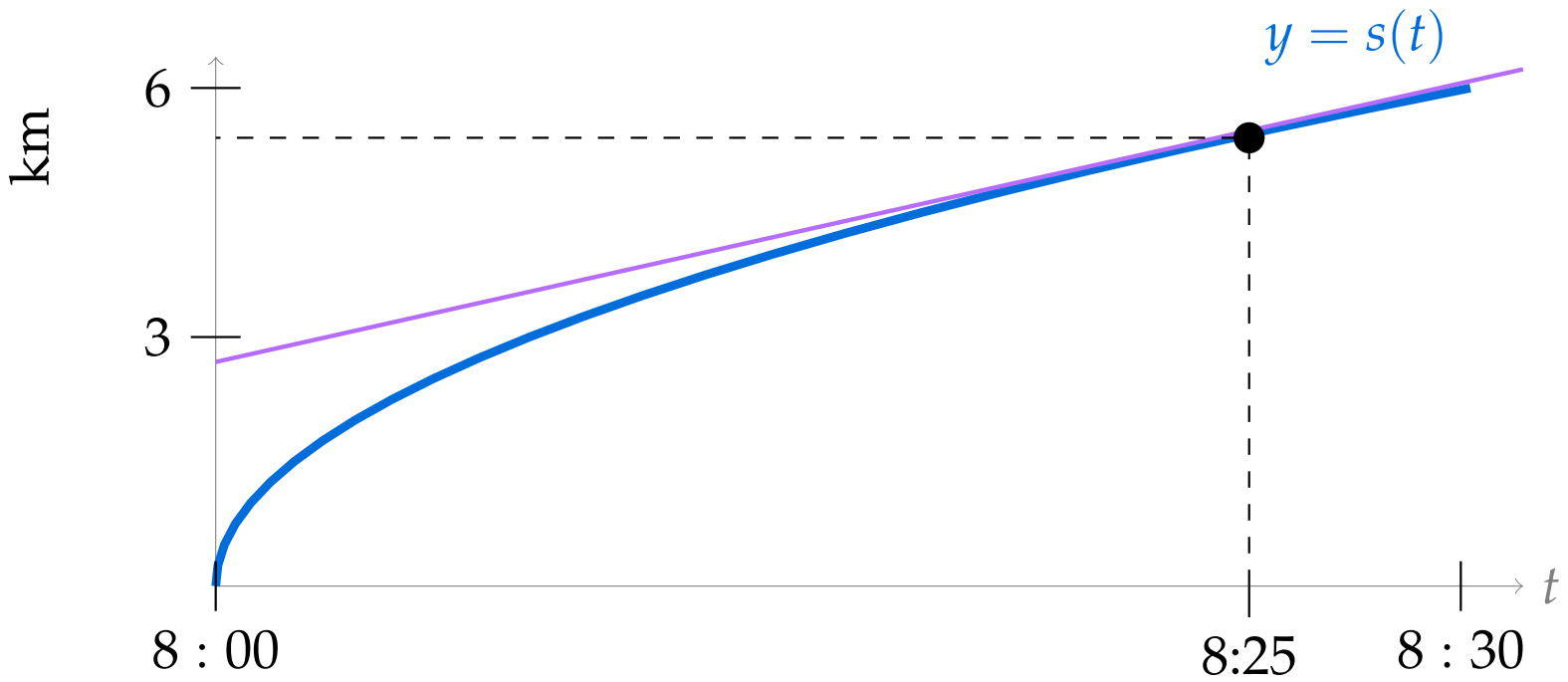
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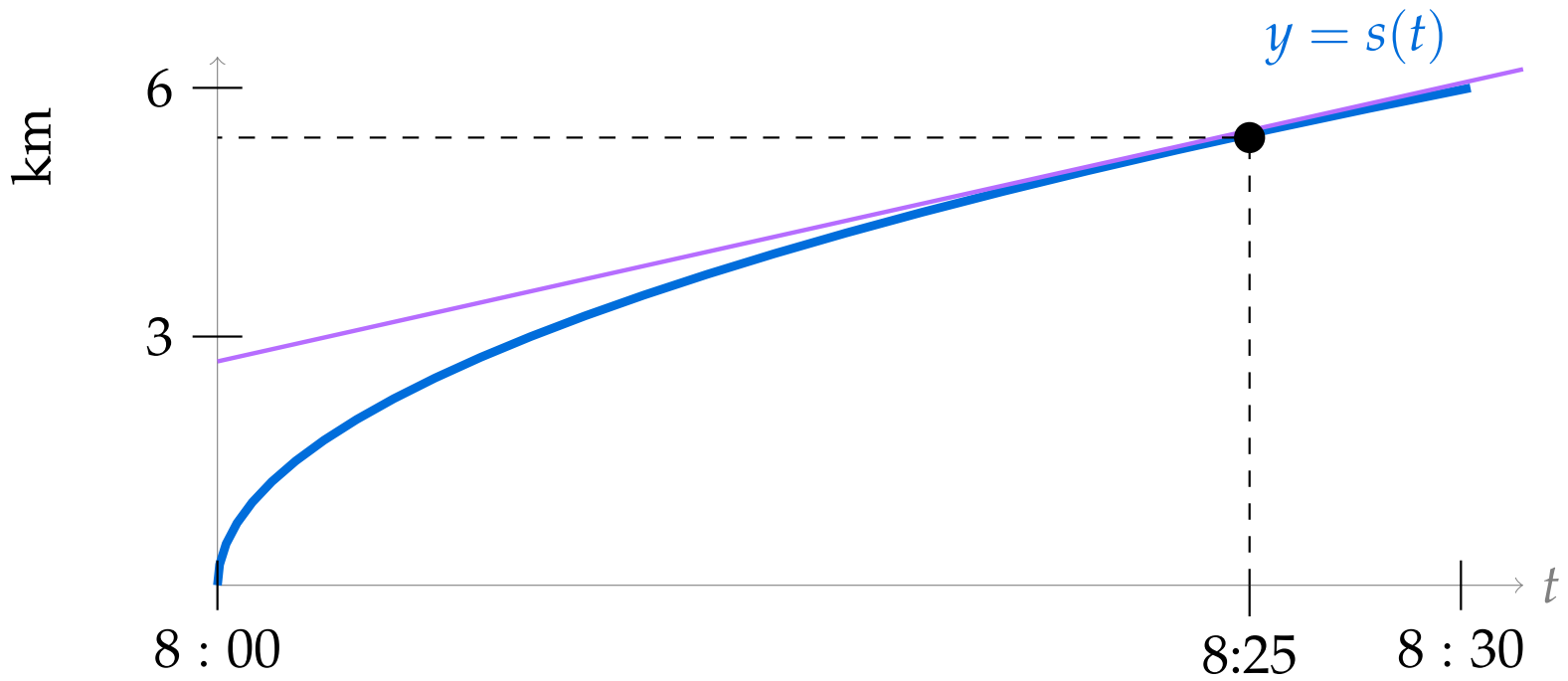
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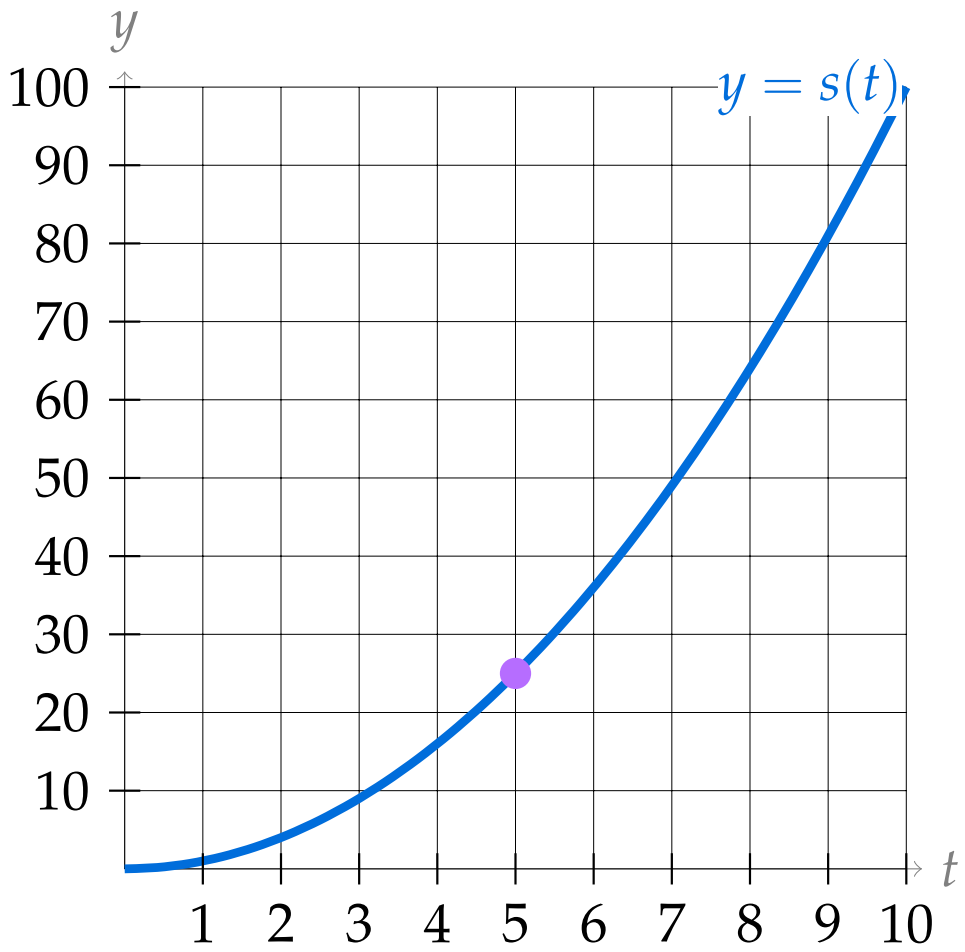
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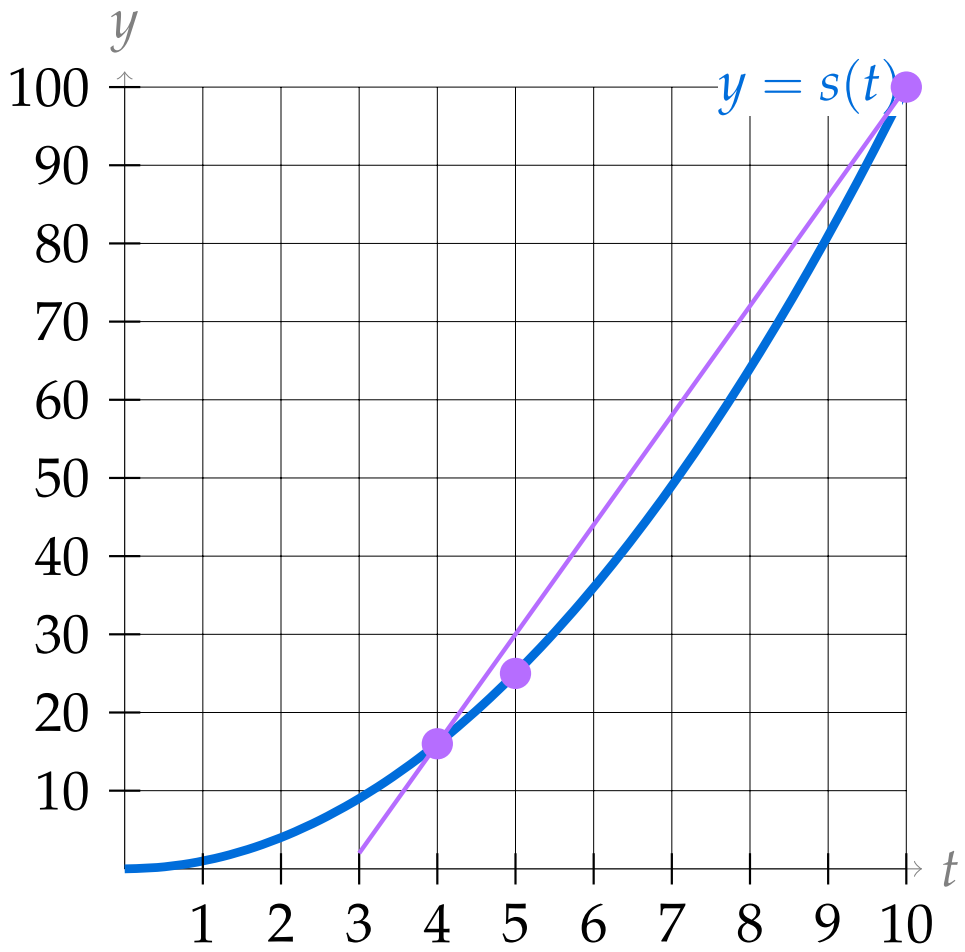
Suppose the distance from the ground s (in metres) of a helium-filled balloon at time t over a 10-second interval is given by $s(t) = t^2$. Try to estimate how fast the balloon is rising when $t = 5$.

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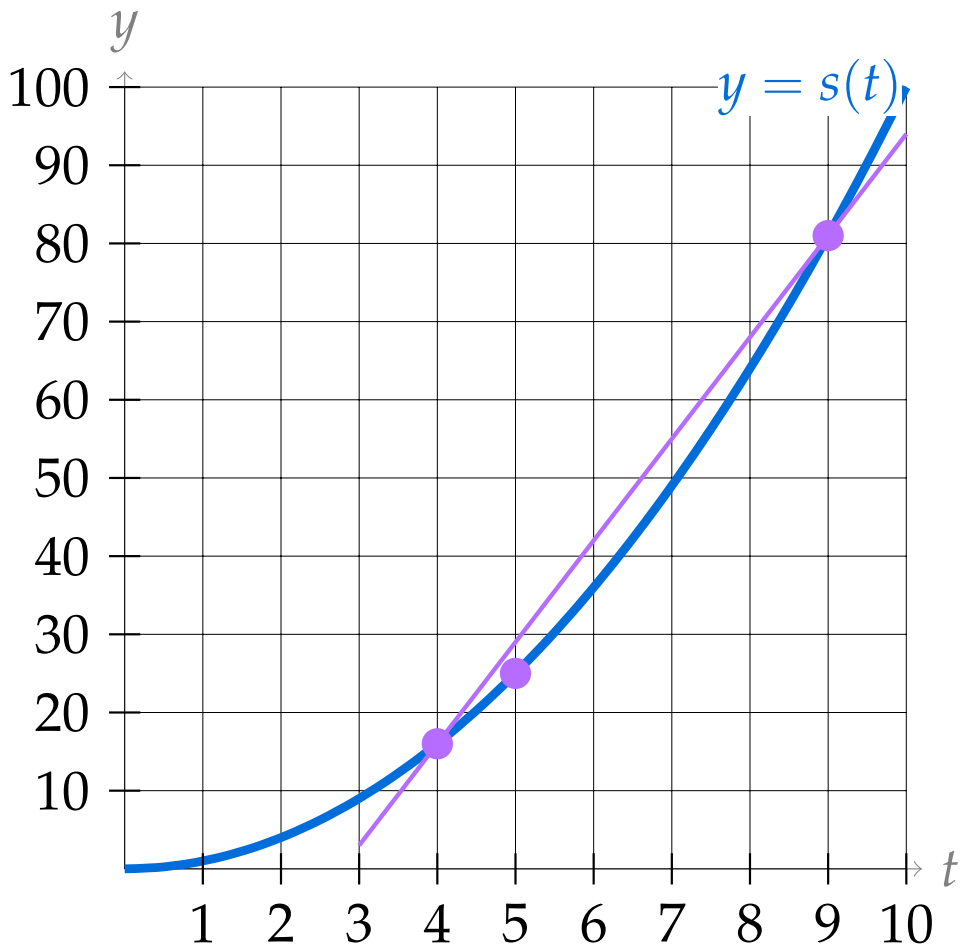
One way: Estimate the slope of the tangent line to the curve

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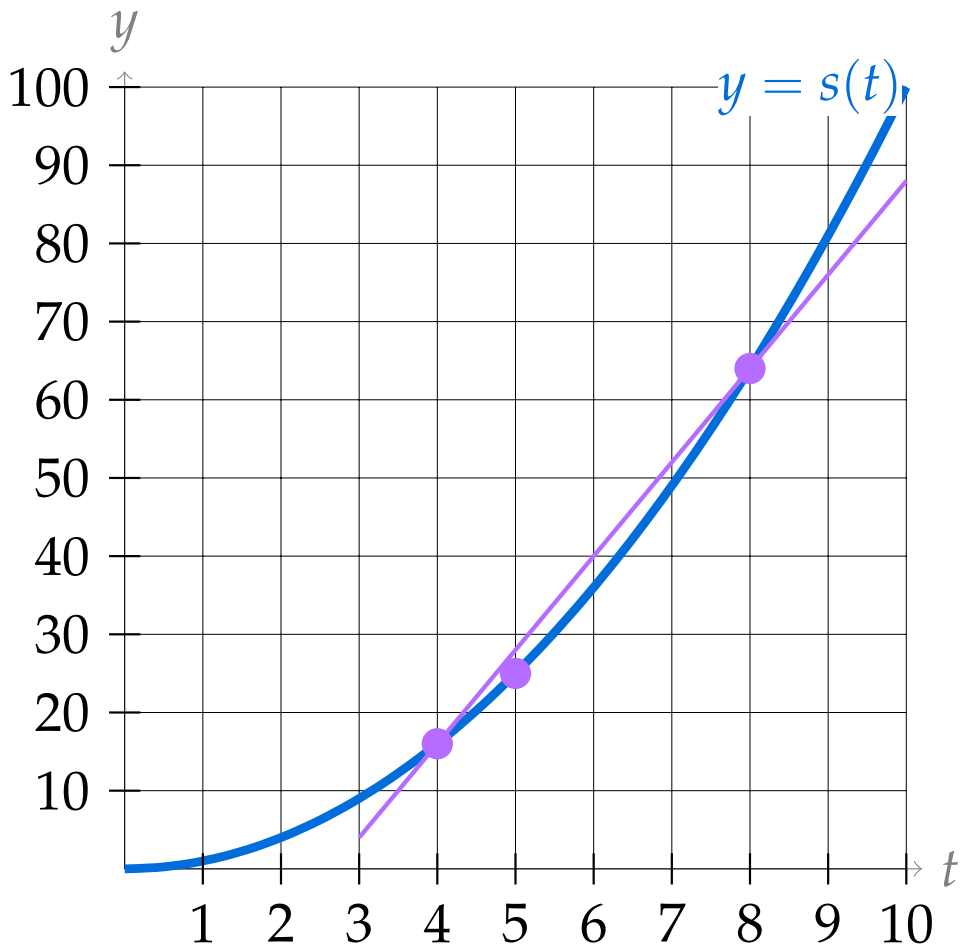
Another way: Calculate average rate of change for intervals around 5 that get smaller and smaller.

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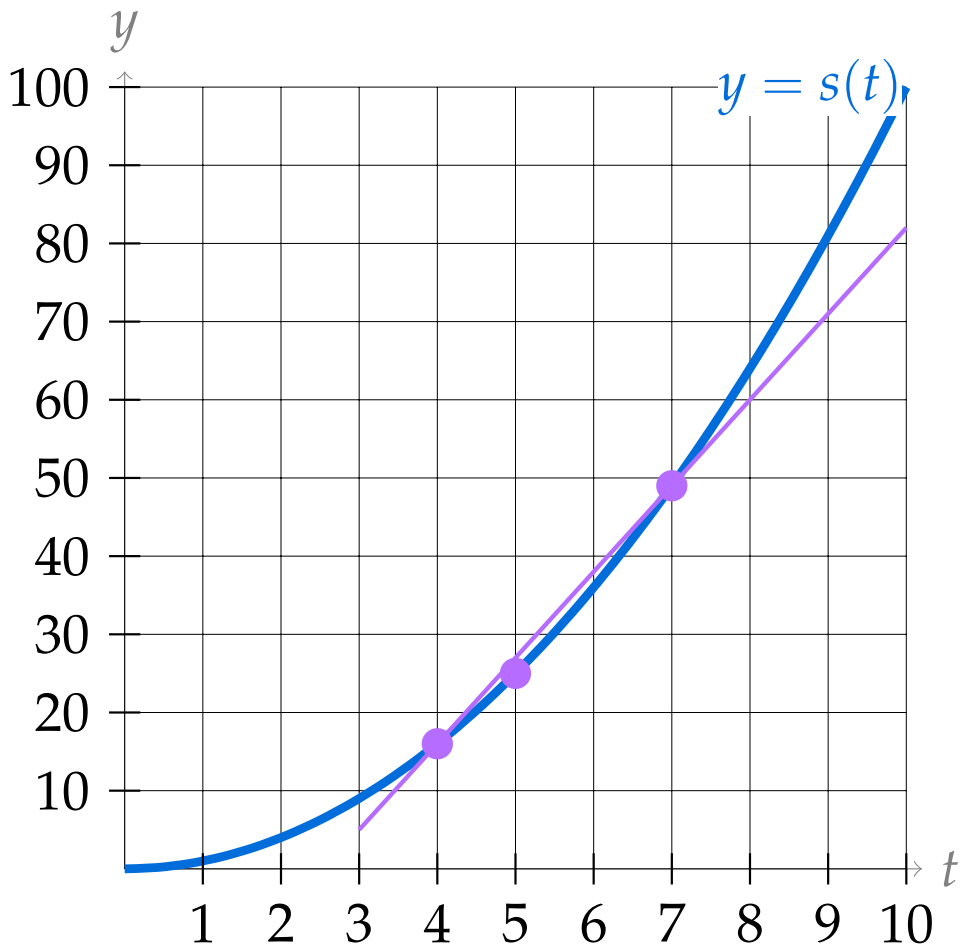
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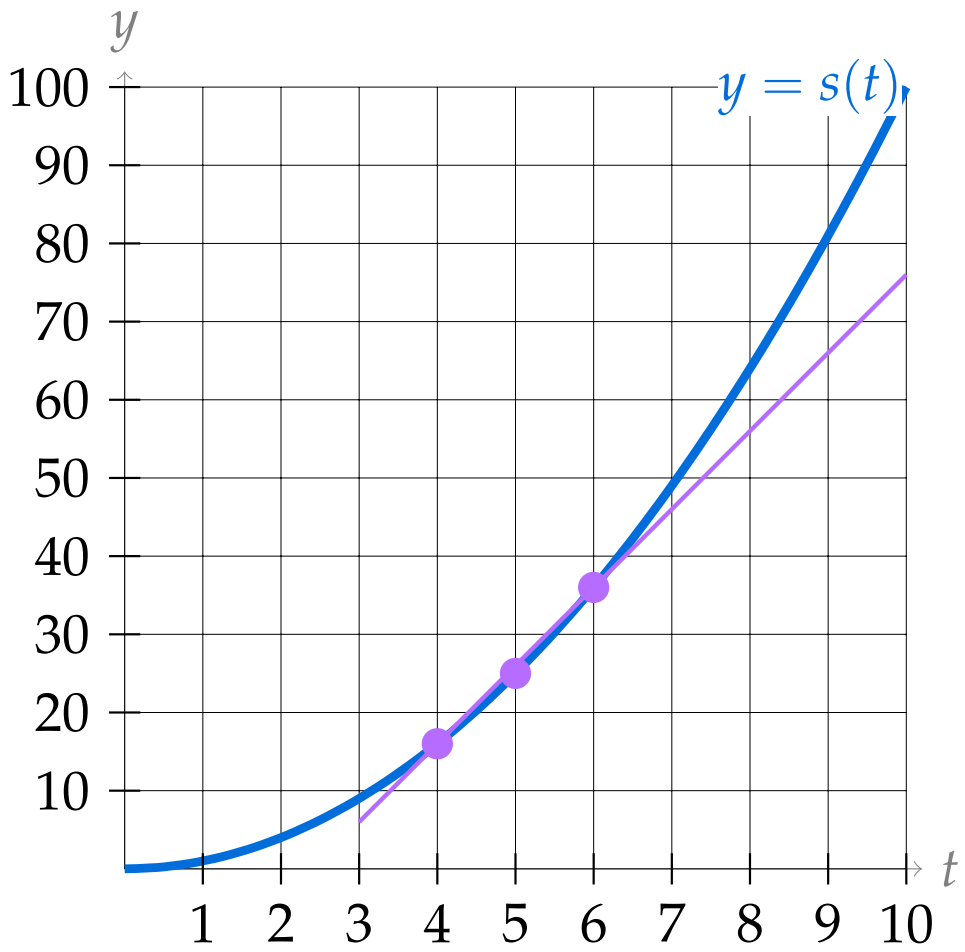
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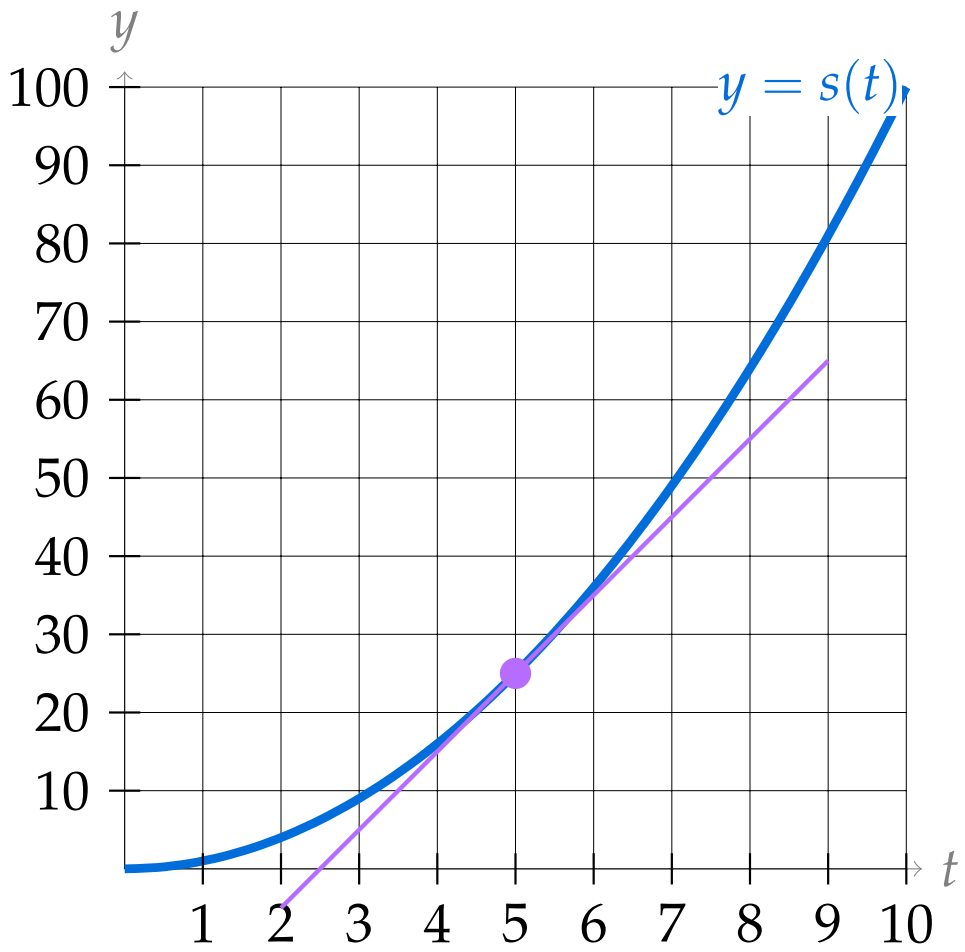
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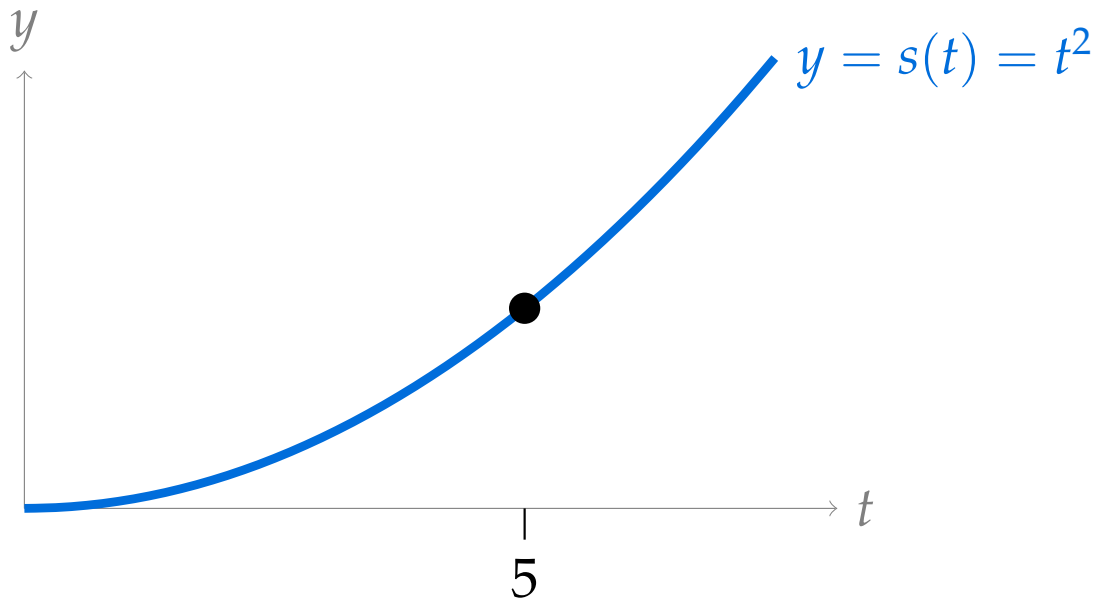
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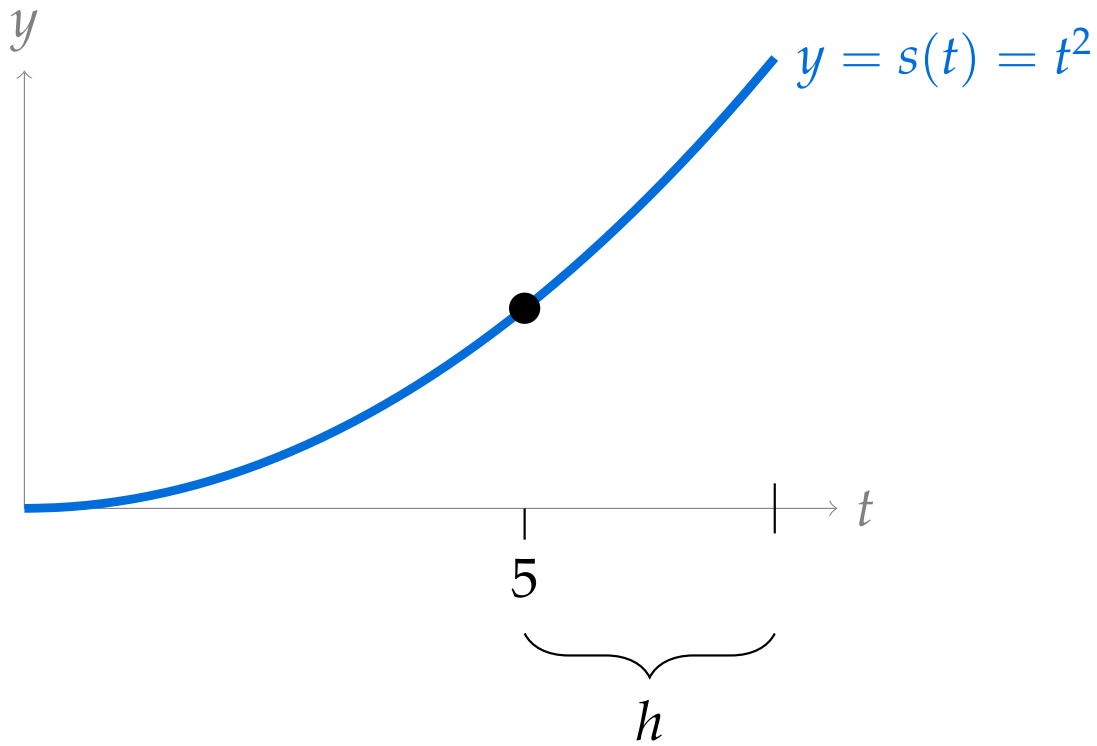


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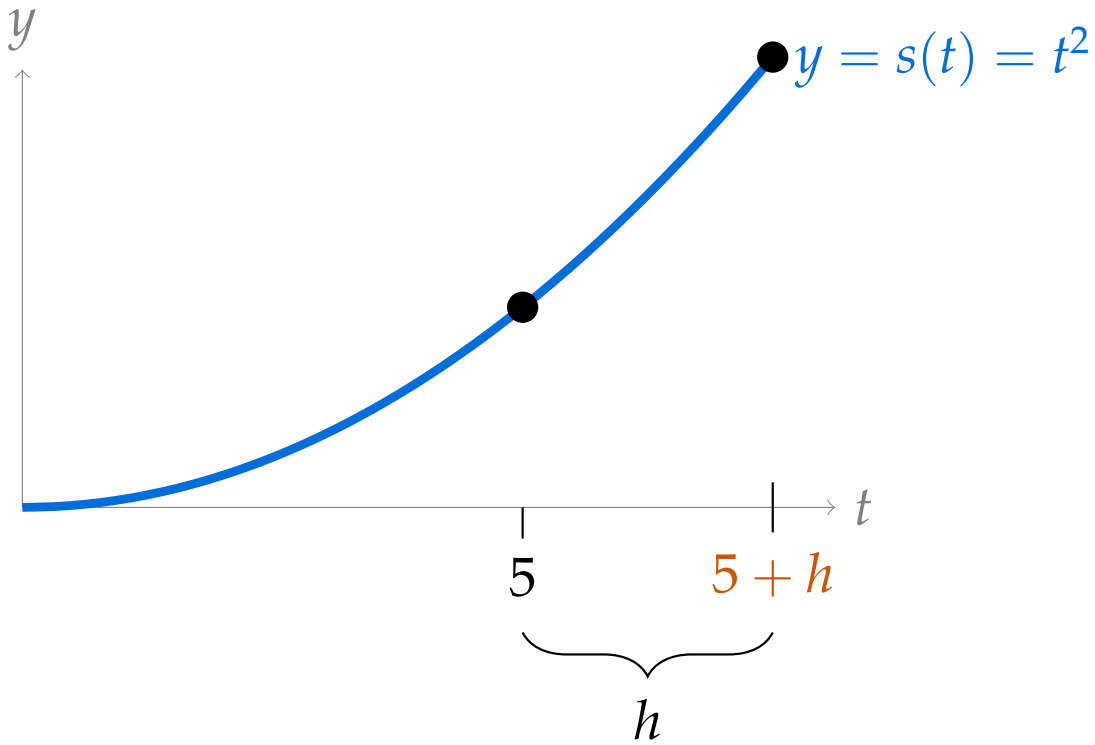
Let's look for an algebraic way of determining the velocity of the balloon when $t = 5$.



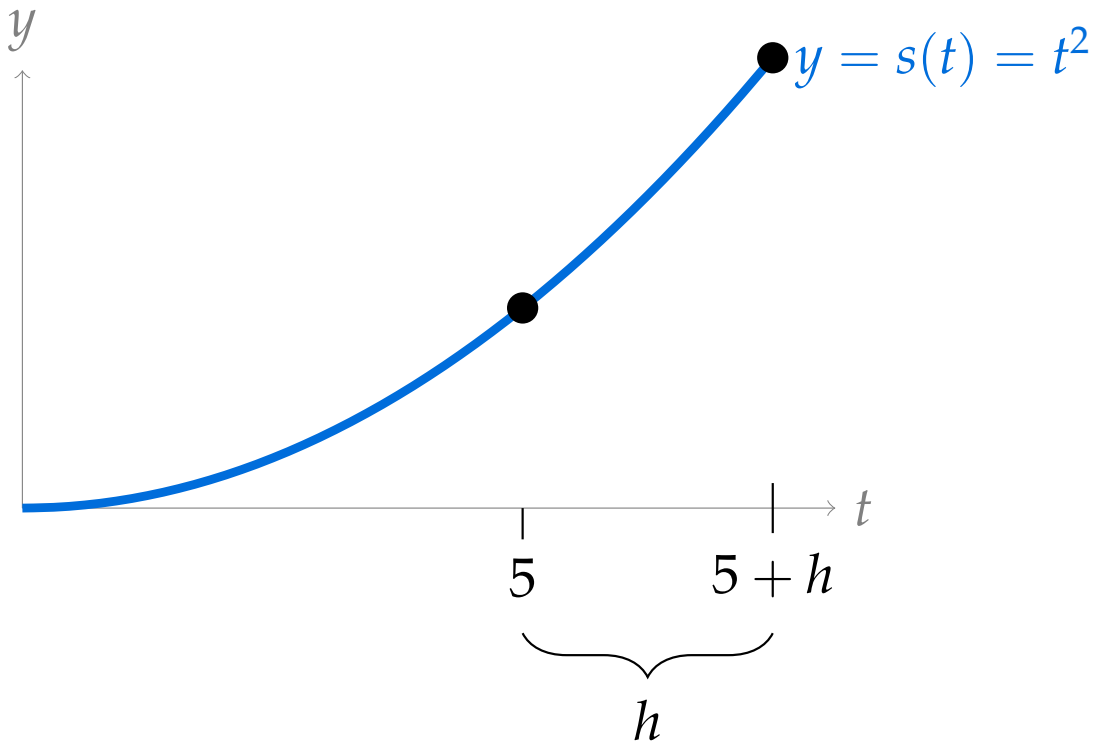
Suppose the interval [5,] has length h. What is the right endpoint of the interval?



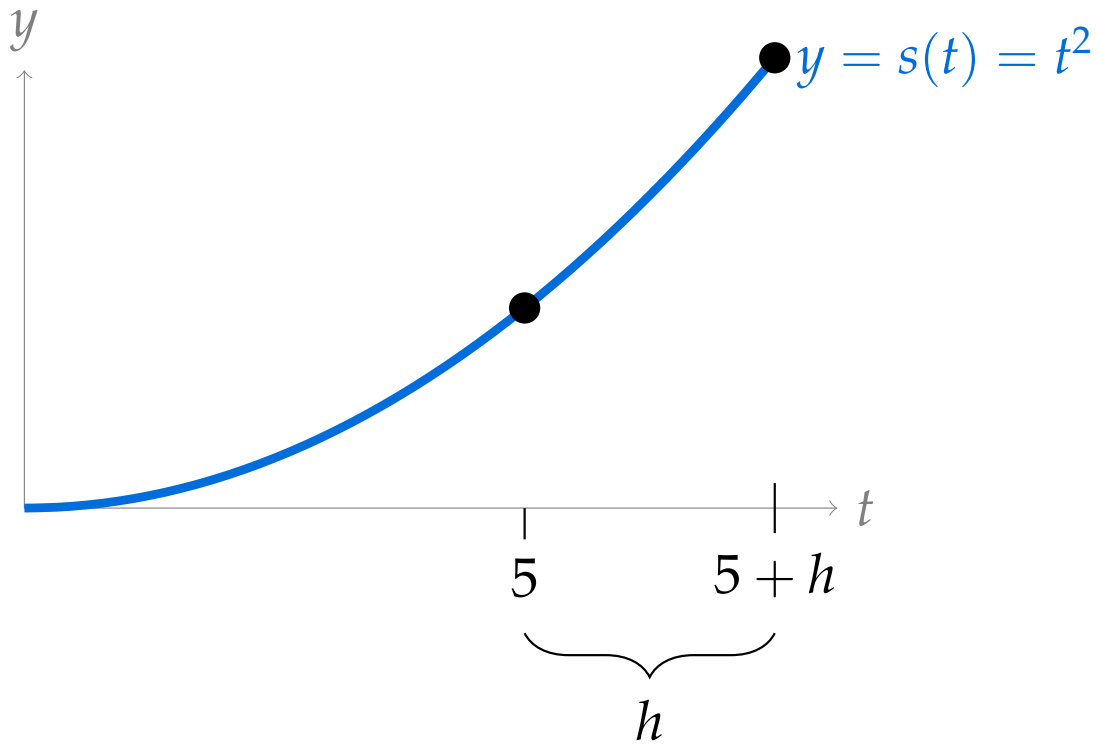
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Write the equation for the average (vertical) velocity from $t = 5$ to $t = 5 + h$.

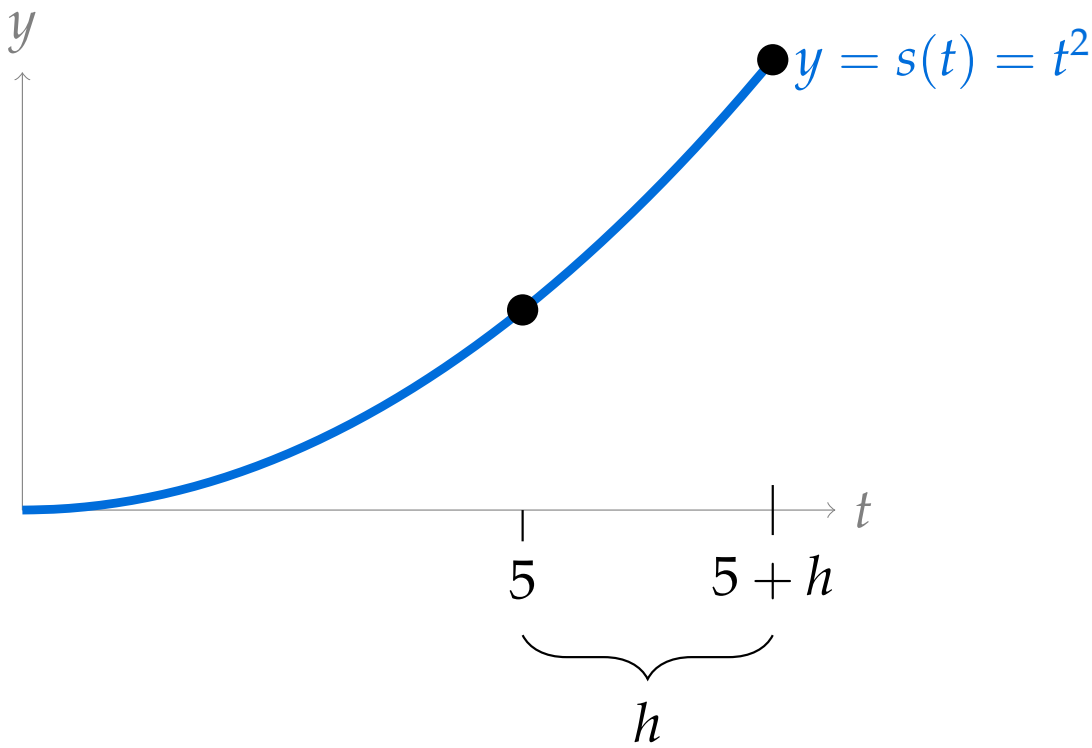


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$$\text{vel} = \frac{\Delta \text{ height}}{\Delta \text{ time}} = \frac{s(5+h) - s(5)}{(5+h) - 5} = \frac{(5+h)^2 - 5^2}{h}$$

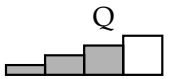
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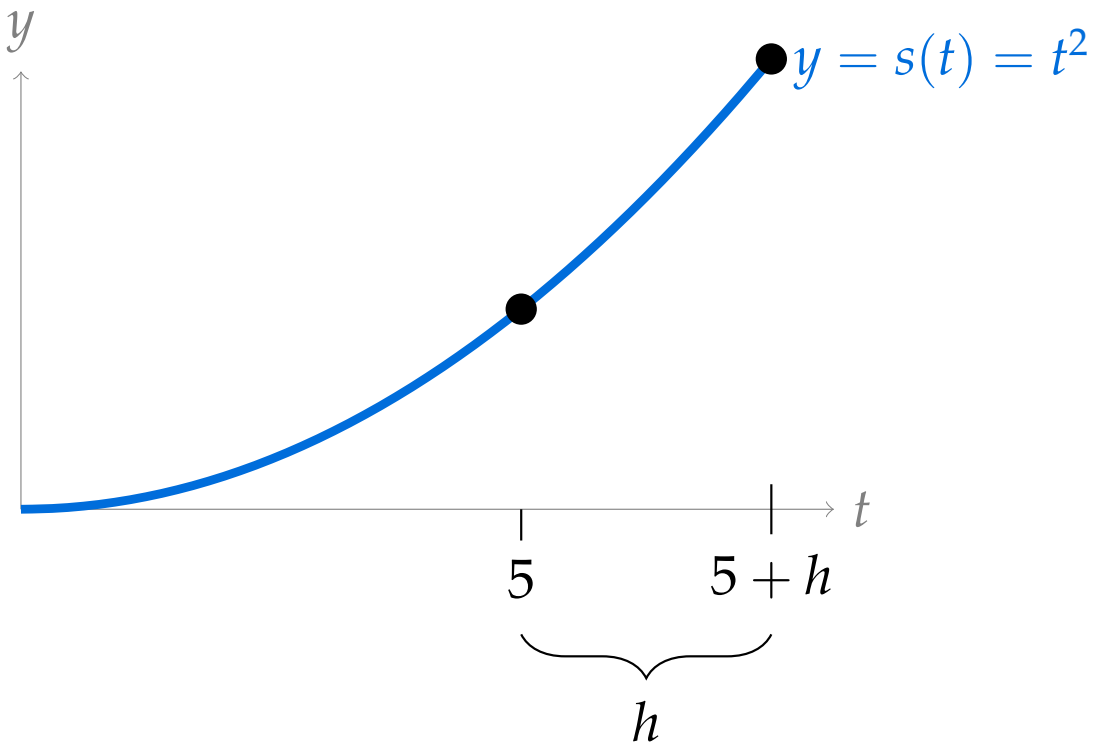
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$$\begin{aligned}\text{vel} &= \frac{\Delta \text{ height}}{\Delta \text{ time}} = \frac{s(5+h)-s(5)}{(5+h)-5} = \frac{(5+h)^2-5^2}{h} \\ &= 10+h \text{ when } h \neq 0\end{aligned}$$

When h is very small,

$$\approx 10$$

What do you think is the slope of the tangent line to the graph when



$t = 5$?

Our First Limit

Average Velocity, $t = 5$ to $t = 5 + h$:

$$\frac{\Delta s}{\Delta t} = \frac{s(5+h) - s(5)}{h}$$

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Limit Notation

We write:

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We say: "The limit as h goes to 0 of $(10 + h)$ is 10."

It means: As h gets extremely close to 0, $(10 + h)$ gets extremely close to 10.

Included Work

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